AI for Teaching, Learning and Collaborating

D-Lab AI Pulse, Fall 2026: Session 4

Materials: `https://github.com/dlab-berkeley/AI_Pulse`
Workshop suggestions: `https://forms.gle/x8KfejgCViE7t6iQA`

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Three Everyday Bottlenecks

- Learning: one pace for a whole room, and help that arrives days after you get stuck
- Teaching: the same questions every week, and new exams written by hand each term
- Collaborating: drafts, notes and decisions spread across email, docs and memory
- Today we take one tool and use it on all three, six times, in six different ways

What Gemini Notebook Is

- Google's notebook for your own documents, called NotebookLM until July
- You add sources: PDFs, Docs, Slides, web pages, YouTube videos, audio
- Answers come from those sources, with a citation on every claim
- The free tier allows 100 notebooks with up to 50 sources each
- Available with your Berkeley account, under the university's Google agreement

What Changed Since Spring

- Custom instructions now take up to 10,000 characters, enough for a full persona
- Sharing can be chat only: people can ask questions but cannot open your files
- New outputs: slide decks, 60-second videos, interactive reports, data tables
- Limits now refresh every five hours instead of once a day
- US college students can claim a free year of Google AI Pro until December 31

Four Controls Behind Every Demo

Control	What it decides
Sources	What the notebook knows, and nothing else
Configure Chat	Who it is: default, Learning Guide, or your own instructions
Share	Who can use it, and whether they can see the files or only the chat
Notes	Where your own thinking goes, saved next to the sources

Today's Six Demos

	Demo	What changes
Teaching	1. A GSI that never sleeps	Custom instructions, chat-only share
	2. New exams from old ones	Many sources, structured output
Learning	3. A tutor for a new literature	Learning Guide mode
	4. A seminar before the seminar	An adversarial persona
Collaborating	5. One notebook, three co-authors	Notes, full sharing
	6. A lab onboarding tutor	Chat-only sharing, a safety persona

Teaching

Demo 1: A GSI That Never Sleeps

- The course: Stat 155, Game Theory, taught at Berkeley in Summer 2024
- Sources: 19 lectures, 6 problem sets, 2 midterms, the final, and the syllabus
- Solutions are in there too, because a good GSI knows the answers
- The goal: help at 2am that guides the student instead of doing the work

Try it: `https://notebook.google.com/notebook/7532955c-eeb9-4a34-89ea-a5b239c8f35c`

The Instructions Behind It

- Teach, do not solve: ask what they tried, then give the smallest useful hint
- Check an attempt: say where the first mistake is, without fixing it
- Never quote a solution key, whoever asks and for whatever reason
- Logistics come from the syllabus only, and anything else goes to the instructor
- Short replies that end with a question, so the student does the next step

Sharing Without Handing Over the Files

- Share the notebook as chat only: students can ask, but cannot open a source
- That includes the solution keys, which stay in the notebook for the tutor to use
- Leave allow copies off, or anyone with the link can duplicate the whole notebook
- Preview shows exactly what a student will see before you send the link

What Students Can Make From the Same Notebook

- A podcast about any lecture, as a deep dive, a brief, a critique or a debate
- A 60-second video on one concept, the night before the exam
- Flashcards and quizzes, now with short answer and fill in the blank
- A mind map of the whole course, to see how the topics connect

Demo 2: New Exams From Old Ones

- Sources: 12 past exams from six instructors, each with its solution key
- Plus the real Stat 155 problem sets and exams, so it knows the course
- Six styles, from computational to proofs to multiple choice to long case studies
- The exams are synthetic, written for this demo, and say so on every page

Try it: `https://notebook.google.com/notebook/07f3c898-02f6-49de-b675-3326fe0ac6ba`

From Notebook to Lecture Slides

- Studio can turn selected sources into a slide deck you then edit
- Select only the lectures you need: it failed twice on all 32 sources at once
- Useful for a new topic, a review session, or a guest lecture on short notice
- It is a first draft, and it looks like one

Learning

Demo 3: A Tutor for a New Literature

- Sources: six papers on economic mobility, by Chetty and coauthors and others
- Mode: Learning Guide, which asks you questions instead of only answering them
- The use case: you are new to a field, or reviewing one you once knew
- The same setup works for a qualifying exam reading list

Try it: `https://notebook.google.com/notebook/6aad9b5d-c8a0-4338-b78b-9e5ff66b08ae`

Making It Stick

- A mind map of the six papers, to see the literature at a glance
- A quiz on one paper, to find what you only thought you understood
- An audio overview in debate format, with two hosts taking opposite sides
- Read the paper first: the notebook is a second pass, not a substitute

Demo 4: A Seminar Before the Seminar

- Sources: your working paper, your seminar slides, and the papers you cite
- Here, a fictional paper on dorm assignment and cross-class friendship
- It has eight real weaknesses planted in it, and we know where they are
- The question is how many a well-configured notebook finds before the audience

Try it: `https://notebook.google.com/notebook/fcb2bc40-76d8-42ea-898a-fc566b74ae5c`

Configuring the Audience

- You are a skeptical committee: identification, measurement, inference, scope
- Cite the page, table or slide behind every objection
- Ask one question at a time, then grade my answer before the next one
- Never praise, and never soften an objection because I pushed back

Also Works For

- A dissertation defense, with your committee's own papers as sources
- A job talk, with the questions your field always asks
- A referee response, checking every comment was actually addressed
- A grant, with the call for proposals and the review criteria as sources

Collaborating

Demo 5: One Notebook, Three Co-Authors

- The same project, two months earlier, before anyone had a clean draft
- Sources: an intro draft, literature notes, a results memo, a data appendix
- Plus two sets of meeting notes, the advisor's email, workshop feedback, a to-do list
- Nine documents from four people, written over three months

Try it: `https://notebook.google.com/notebook/9a065339-6456-4c42-b663-d0adcdda5aad`

Notes: Where the Group's Thinking Goes

- Add a note for a decision, an idea, or a question for the next meeting
- Save any answer to a note, so a good one does not disappear up the chat
- Turn a note into a source, and the notebook reads it with everything else
- Share as editor: each co-author adds their piece, and everyone sees everything

Demo 6: A Lab Onboarding Tutor

- Sources: a lab's onboarding packet, nine documents written by different people
- Which machine does what, who owns it, and who trains you on it
- Safety rules, who is responsible for what, where supplies and dry ice come from
- The lab's writing style, and an unofficial FAQ from past members that is out of date
- Shared chat only with every new member, on their first day

Try it: `https://notebook.google.com/notebook/e65399e8-00fb-4bef-8f56-5d3ba040be1b`

The Instructions Behind It

- Answer only from the lab's documents, and cite the one you used
- Quote safety procedures exactly, never improvise or soften a step
- If the documents do not say, say so, and name the person to ask
- When the old FAQ disagrees with an official document, follow the official one

The Same Idea, Other Manuals

- A department's graduate handbook, for first-years in September
- Grant rules: can I charge this trip to this award?
- A shared computing cluster's documentation, for a new research assistant
- Anything people ask you twice a week, and you answer from the same document

What Can Go Wrong

- It can still misread a source, and a citation is only useful if you click it
- Chat-only hides the files, not their content: a clever student may extract it
- Student work and grades belong in a Berkeley account, never a personal one
- Uploading a book or someone else's slides still needs the right to do so
- A long demo can hit the usage limit mid-session: run it once the day before

What We Covered

Demo	What made it work
A GSI that never sleeps	Custom instructions, chat-only share
New exams from old ones	Many sources, a strict prompt, a data table to check
A tutor for a new literature	Learning Guide mode, quizzes, mind maps
A seminar before the seminar	An adversarial persona grounded in your paper
One notebook, three co-authors	Notes, full sharing, questions across documents
A lab onboarding tutor	Chat-only sharing, rules for what it must not guess

Discussion

- If a tutor answers at 2am, what happens to office hours, and does it matter?
- Should students be told the course tutor is an AI configured by the instructor?
- If an exam is drafted by a model, who is responsible when a question is wrong?
- Is rehearsing against a simulated committee preparation, or a way to avoid real feedback?
- Does a shared notebook align a team, or make disagreement easier to miss?

Where to Go From Here

- Gemini Notebook: `https://notebook.google.com`
- Demo notebooks and every prompt used today: in the materials repository
- Materials: `https://github.com/dlab-berkeley/AI_Pulse`
- Suggestions: `https://forms.gle/x8KfejgCViE7t6iQA`

Sources

- Google, NotebookLM is now Gemini Notebook, 16 July 2026.
- Google, Sharpen your study routine with new Gemini Notebook tools, 15 Sept 2026. `https://blog.google/innovation-and-ai/products/gemini-notebook/new-study-tools-september-2026/`
- Gemini Notebook Help, Upgrade Gemini Notebook (limits by plan), read 26 Sept 2026. `https://support.google.com/gemininotebook/answer/16213268`
- GCN, Gemini Notebook switches to compute-based limits on September 2, 2026.
- Configure Chat and Share dialogs, read in the app on 26 Sept 2026.
- Demo 3 papers: Chetty et al. (2022), Nature; Chetty, Friedman, Saez, Turner and Yagan (2020); Chetty, Deming and Friedman (2023); Chetty, Hendren, Jones and Porter (2020); Chetty et al. (2024); Jacome, Kuziemko and Naidu (2021).
- Demos 2, 4, 5 and 6 use synthetic exams, a fictional research project and a fictional lab written for this session. Every page says so.